

Sai Kiran Mayakuntla

Quantitative Analytics Specialist
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EDUCATION

- **PhD + M.Sc. (Engg)** Aug 2013 - July 2019
Indian Institute of Science (IISc) - Dept. of Civil Engineering; GPA: 6.4/8.0
Thesis: Macroscopic modelling of heterogeneous, disordered road traffic flow
Bengaluru, India
- **B.Tech.** Aug 2009 - Apr 2013
JNTUA - Dept. of Civil Engineering; Overall score: 75%
Anantapur, India

SKILLS

- **Finance:** Financial markets, derivative pricing, counterparty credit risk, market risk, and model risk management.
- **Mathematics:** Stochastic calculus, probability theory, statistics, optimization techniques and game theory.
- **Programming:** Python, Julia; Familiarity with object-oriented and functional programming concepts.

EXPERIENCE

- **Quantitative Analytics Specialist** Wells Fargo ISPL, Hyderabad, India
Validation of Counterparty Credit Risk (CCR) models. Jan 2025 - Present
 - Performed model risk assessment of Potential Future Exposure (PFE) and Credit/Debit Valuation Adjustment (XVA) models, covering their conceptual soundness, performance testing, implementation accuracy, and regulatory compliance.
 - Performed independent testing (benchmarking, sensitivity analysis, convergence tests, etc.) using Python and Excel VBA.
 - Contributed to the team wiki documentation on the use of internal quantitative libraries.
 - Coordinated with stakeholders across teams to drive project execution and meet deadlines.
 - Worked on multiple projects spanning different stages of the model lifecycle, from initial validation to periodic and focused reviews.
- **BA4 Traded Models Validation AM** Barclays GSCPL, Noida, India
Validation of Market Risk and CCR models. Nov 2022 - Dec 2024
 - Performed in-depth methodological reviews, input data quality assessments, and regulatory compliance checks for multiple CCR and Market Risk models as part of the Traded Model Risk team.
 - Conducted initial validation of an Agency Mortgage-Backed Securities (MBS) exposure calculation model feeding into the CCR engine, and validated changes to the VaR model.
 - Assessed model performance with independent benchmark testing and outcomes analysis using Python.
- **Postdoctoral researcher** Universidad de Chile, Santiago, Chile
Worked on various problems in transportation networks analysis April 2021 - Oct 2022
 - Developed route choice models in transportation networks based on Game theoretic principles, specifically Nash Equilibrium.
 - Developed Julia code-base to simulate city-level road traffic networks and implement custom optimization algorithms.
- **Technical consultant** Fluentem India Inc., Bengaluru
Advised on the development of a micro-simulation software for Indian road traffic Nov 2020 - March 2021
Researched various aspects relevant to its development and illustrated the ideas using Julia programming language.
- **Senior research fellow** Indian Institute of Science, Bengaluru
Studied the impact of utility restoration works on Bengaluru city's road traffic Oct 2019 - Feb 2020
Used simulation techniques to study the impact of utility restoration works on the road traffic in Bengaluru city.
- **Graduate research assistant** Indian Institute of Science, Bengaluru
Completed my M.Tech. course work and worked on my PhD thesis Aug 2013 - July 2019
My thesis work included modelling traffic flow on Indian roads and an application of the model to vehicle route assignment.

ACHIEVEMENTS

- Successfully cleared Part 1 of the Financial Risk Manager (FRM) exam, administered by GARP, achieving top quartile scores across all four subject areas.
- Authored 8 peer-reviewed research papers in Transportation Engineering, published in leading international journals such as Transportation Research Part D and the Journal of Transportation Engineering Part A.
- Secured All India Rank 118 in the Graduate Aptitude Test in Engineering (GATE) for Civil Engineering, 2012.

OTHER INTERESTS

- I often publish blog posts on financial mathematics and selected Jane Street puzzles on LinkedIn.